SysML v2 structural view from the canonical IR (template path)

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Scope and estimate. Project the structural part of a SysML-profiled HyMeKo model (blocks with specialization, ports, composition with multiplicity, n-ary item-flow connectors, deployment) to SysML v2 text through the existing query+template path. User decision 2026-10-06: template path with a minimal engine extension, type-level IBD (connection defs), no nested part usages. ETA 60–90 minutes. No benchmark, no campaign.

Ownership. Core: compiler and IR (read only, `Ir::parent`). Query owns template semantics: three new expressions in `rewrite/template.rs` reusing `bases::bases`. Formats owns the transform registration (`SysmlStructureTransform`, template-only like `RequirementsSysmlTransform`). `transforms/sysml_structure/` owns query and template text. Vocabulary: `data/sysml/meta_sysml_structu` (imports the unchanged paper witness vocabulary). The requirements transform and its paper contract stay untouched.

Engine API (template expressions). {{`owner`}}: name of the current match's parent declaration. {{`bindowner:s:i`}}: parent name of the i-th binding with sign s (ports are children of their block, so `bindowner`+`bind` yields `Block_port`). {{`bases:L`}}: comma-separated direct bases (`BasePolicy::All`, authored order) that are themselves matches of query label `L`; vocabulary metaclasses are therefore excluded. {{`#if ...`}} already tests "expression non-empty", so no new block type is needed.

Files. Query: `src/rewrite/template.rs` (+ unit tests). Formats: `src/transforms.rs`, `src/lib.rs`. Transforms: `sysml_structure/queries.hymeko`, `template.sysml`. Data: `sysml/meta_sysml_structure.hyme` `sysml/sysml_structure_witness.hymeko`. Tests: `hymeko_query/tests/test_transform_ecosystem.rs` (new module). SYMBOLS.md. This plan; report `reports/2026-10-06-sysml-structure-template.md` (gitignored, archived).

Tests. Unit: each expression on an in-memory model, with counter-tests (a base outside the label is not emitted; a root-level match has an empty owner; an out-of-range binding is empty). Integration: render the witness and assert `part def B :> A`, port defs named `Owner_port`, composition multiplicities from a shared reference and from constant arithmetic, n-ary connector ends plus all neutral items, deployment allocation; and that the registry still lists the existing transforms. Counter-test: removing the `bases:` filter label makes the metaclass appear, proving the assertion fires for the right reason.

Budget and risks. Expressions are O(bases $\times$ label size) per match; label membership is a `HashSet` built per expression evaluation. SysML v2 output is not validated by a SysML v2 tool in this task (stated as a limitation). Type-level connectors and composition-as-connection-def are a projection, not the usage tree.

State and rollback. Fetch done by the user (2026-10-06). The checkout is on `feat/material-lobes`; touched tracked files are byte-identical to `origin/main`. No git write operations from the Linux sandbox (a stray `index.lock` incident occurred and was cleaned); the user branches and commits on Windows. Verification in a sandbox build of `git archive HEAD` overlaid with the changed files and the pinned dependencies. Rollback: delete the new files, revert the named edits.

